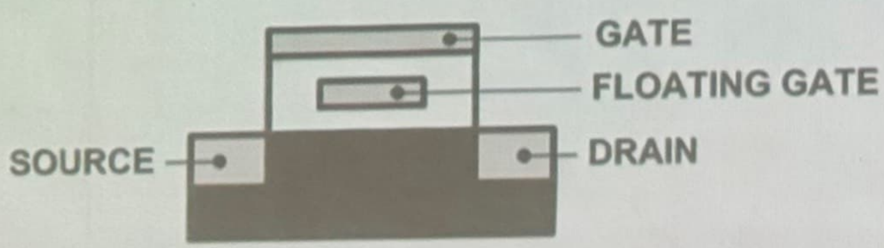
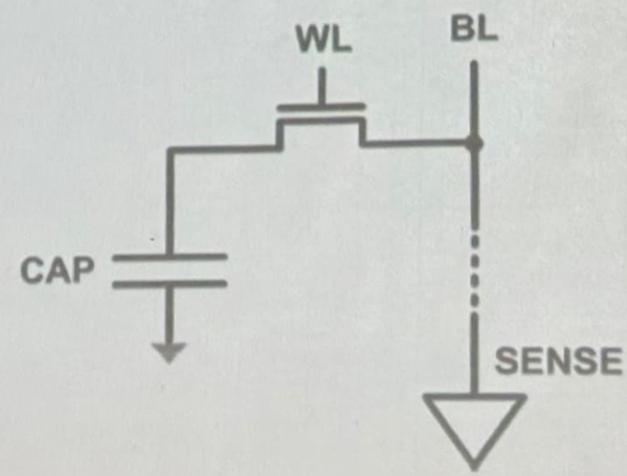
Computer Architecture

Emerging Memory Technologies

Limits of Charge Memory

- Difficult charge placement and control
  - Flash: floating gate charge
  - DRAM: capacitor charge, transistor leakage
- Reliable sensing becomes difficult as charge storage unit size reduces





Solution 1: New Memory Architectures

- Overcome memory shortcomings with
  - Memory-centric system design
  - Novel memory architectures, interfaces, functions
  - Better waste management (efficient utilization)

- Key issues to tackle
  - Enable reliability at low cost → high capacity
  - Reduce energy
  - Reduce latency
  - Improve bandwidth
  - Reduce waste (capacity, bandwidth, latency)
  - Enable computation close to data

Solution 1: New Memory Architectures

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- Kim+, "A Case for Exploiting Subarray-Level Parallelism in DRAM," ISCA 2012.
- Lee+, "Tiered-Latency DRAM: A Low Latency and Low Cost DRAM Architecture," HPCA 2013.
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- Seshadri+, "RowClone: Fast and Efficient In-DRAM Copy and Initialization of Bulk Data," MICRO 2013.
- Pekhimenko+, "Linearly Compressed Pages: A Main Memory Compression Framework," MICRO 2013.
- Chang+, "Improving DRAM Performance by Parallelizing Refreshes with Accesses," HPCA 2014.
- Khan+, "The Efficacy of Error Mitigation Techniques for DRAM Retention Failures: A Comparative Experimental Study," SIGMETRICS 2014.
- Luo+, "Characterizing Application Memory Error Vulnerability to Optimize Data Center Cost," DSN 2014.
- Kim+, "Flipping Bits in Memory Without Accessing Them: An Experimental Study of DRAM Disturbance Errors," ISCA 2014.
- Lee+, "Adaptive-Latency DRAM: Optimizing DRAM Timing for the Common-Case," HPCA 2015.
- Qureshi+, "AVATAR: A Variable-Retention-Time (VRT) Aware Refresh for DRAM Systems," DSN 2015.
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- Kim+, "Ramulator: A Fast and Extensible DRAM Simulator," IEEE CAL 2015.
- Seshadri+, "Fast Bulk Bitwise AND and OR in DRAM," IEEE CAL 2015.
- Ahn+, "A Scalable Processing-in-Memory Accelerator for Parallel Graph Processing," ISCA 2015.
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- Seshadri+, "Gather-Scatter DRAM: In-DRAM Address Translation to Improve the Spatial Locality of Non-unit Strided Accesses," MICRO 2015.
- Lee+, "Simultaneous Multi-Layer Access: Improving 3D-Stacked Memory Bandwidth at Low Cost," TACO 2016.
- Hassan+, "ChargeCache: Reducing DRAM Latency by Exploiting Row Access Locality," HPCA 2016.
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- Hsieh+, "Transparent Offloading and Mapping (TOM): Enabling Programmer-Transparent Near-Data Processing in GPU Systems," ISCA 2016.
- Hashemi+, "Accelerating Dependent Cache Misses with an Enhanced Memory Controller," ISCA 2016.
- Boroumand+, "LazyPIM: An Efficient Cache Coherence Mechanism for Processing-in-Memory," IEEE CAL 2016.
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- Khan+, "A Case for Memory Content-Based Detection and Mitigation of Data-Dependent Failures in DRAM"," IEEE CAL 2016.
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- Seshadri and Mutlu, "Simple Operations in Memory to Reduce Data Movement," ADCOM 2017.
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- Boroumand+, "Google Workloads for Consumer Devices: Mitigating Data Movement Bottlenecks," ASPLOS 2018.
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- Wang+, "Reducing DRAM Latency via Charge-Level-Aware Look-Ahead Partial Restoration," MICRO 2018.
- Kim+, "D-RaNGe: Using Commodity DRAM Devices to Generate True Random Numbers with Low Latency and High Throughput," HPCA 2019.
- Singh+, "NAPEL: Near-Memory Computing Application Performance Prediction via Ensemble Learning," DAC 2019.
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- Kim+, "Revisiting RowHammer: An Experimental Analysis of Modern Devices and Mitigation Techniques," ISCA 2020.
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- Patel+, "Bit-Exact ECC Recovery (BEER): Determining DRAM On-Die ECC Functions by Exploiting DRAM Data Retentio Characteristics," MICRO 2020.
- Avoid DRAM:
    - Seshadri+, "The Evicted-Address Filter: A Unified Mechanism to Address Both Cache Pollution and Thrashing," PACT 2012.
    - Pekhimenko+, "Base-Delta-Immediate Compression: Practical Data Compression for On-Chip Caches," PACT 2012.
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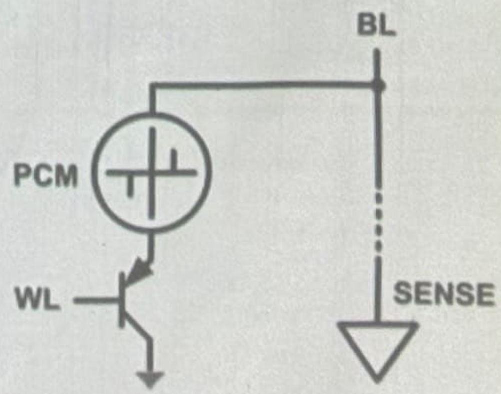
Solution 2: Emerging Memory Technologies

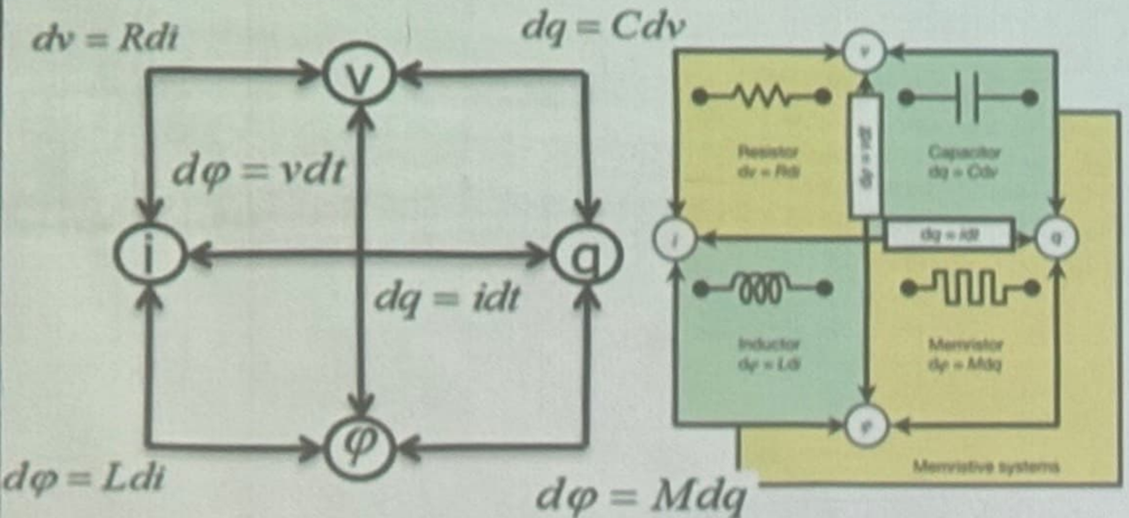
- Some emerging **resistive** memory technologies seem more scalable than DRAM (and they are non-volatile)

- Example: Phase Change Memory
  - Data stored by changing phase of material
  - Data read by detecting material's resistance
  - Expected to scale to 9nm (2022 [ITRS 2009])
  - Prototyped at 20nm (Raoux+, IBM JRD 2008)
  - Expected to be denser than DRAM: can store multiple bits/cell

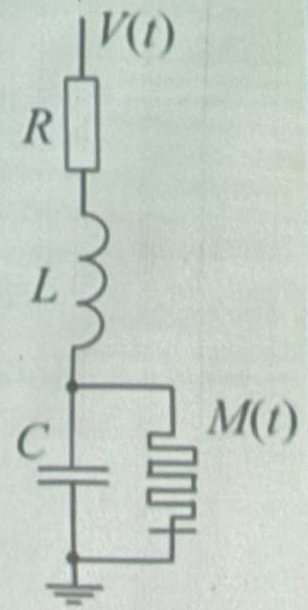


Missing Circuit Element

- Hypothesized by Chua in 1971



Theoretical circuit used for emulating learning responses in *P. polycephalum*. The inductor and capacitor simulate biological oscillations, the resistor attenuates the response and the memristor alters the reaction from the RLC circuit. The input and output voltages represent the stimuli and the response respectively.



- Use of multiple circuits in parallel would allow for simultaneous learning and advanced signal processing.
- Artificial intelligence implemented by various hardware elements instead of elaborate software systems.

Memristor-The missing circuit element Chua 1971 IEEE TCAS

Solution 2: Emerging Memory Technologies

- Lee+, "Architecting Phase Change Memory as a Scalable DRAM Alternative," ISCA'09, CACM'10, IEEE Micro'10.
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- Meza+, "A Case for Efficient Hardware-Software Cooperative Management of Storage and Memory," WEED 2013.
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- Yoon, Meza+, "Efficient Data Mapping and Buffering Techniques for Multi-Level Cell Phase-Change Memories," TACO 2014.
- Ren+, "ThyNVM: Enabling Software-Transparent Crash Consistency in Persistent Memory Systems," MICRO 2015.
- Chauhan+, "NVMove: Helping Programmers Move to Byte-Based Persistence," INFLOW 2016.
- Li+, "Utility-Based Hybrid Memory Management," CLUSTER 2017.
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- Sadrosadati+. "LTRF: Enabling High-Capacity Register Files for GPUs via Hardware/Software Cooperative Register Prefetching," ASPLOS 2018.
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- Wang+, "Panthera: Holistic Memory Management for Big Data Processing over Hybrid Memories," PLDI 2019.
- Song+, "Enabling and Exploiting Partition-Level Parallelism (PALP) in Phase Change Memories," CASES 2019.
- Liu+, "Binary Star: Coordinated Reliability in Heterogeneous Memory Systems for High Performance and Scalability," MICRO'19.
- Song+, "Improving Phase Change Memory Performance with Data Content Aware Access," ISMM 2020.

Charge vs. Resistive Memories

- Charge Memory (e.g., DRAM, Flash)
  - Write data by capturing charge Q
  - Read data by detecting voltage V

- Resistive Memory (e.g., PCM, STT-MRAM, memristors)
  - Write data by pulsing current dQ/dt
  - Read data by detecting resistance R

Promising Resistive Memory Technologies

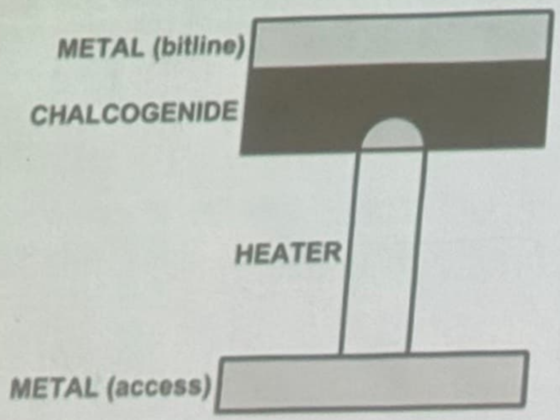
- PCM
  - Inject current to change material phase
  - Resistance determined by phase

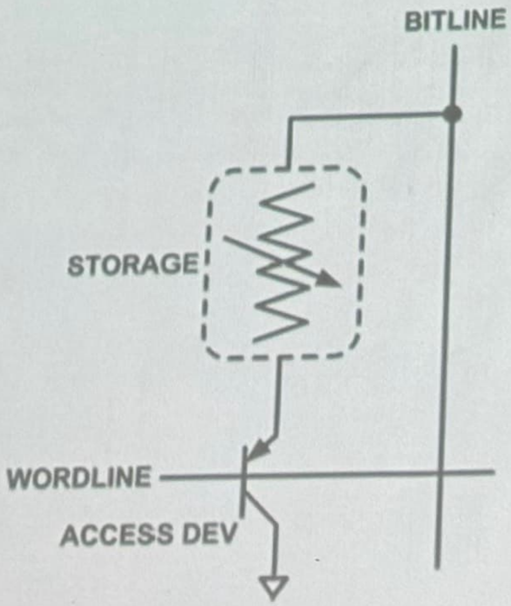
- STT-MRAM
  - Inject current to change magnet polarity
  - Resistance determined by polarity

- Memristors/RRAM/ReRAM
  - Inject current to change atomic structure
  - Resistance determined by atom distance

What is Phase Change Memory?

- Phase change material (chalcogenide glass) exists in two states:
  - Amorphous: Low optical reflexivity and high electrical resistivity
  - Crystalline: High optical reflexivity and low electrical resistivity

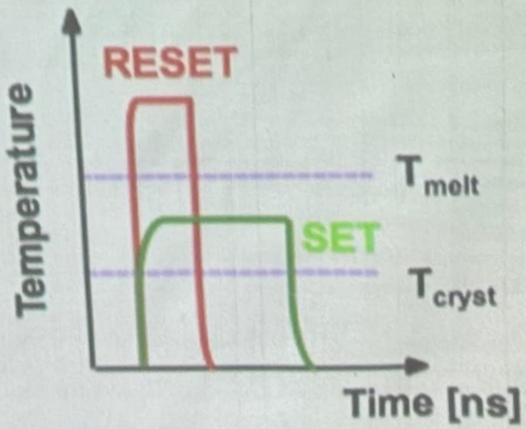




PCM is resistive memory: High resistance (0), Low resista
PCM cell can be switched between states reliably and

How Does PCM Work?

- Write: change phase via current injection
  - SET: sustained current to heat cell above T*cryst*
  - RESET: cell heated above T*melt* and quenched
- Read: detect phase via material resistance
  - amorphous/crystalline



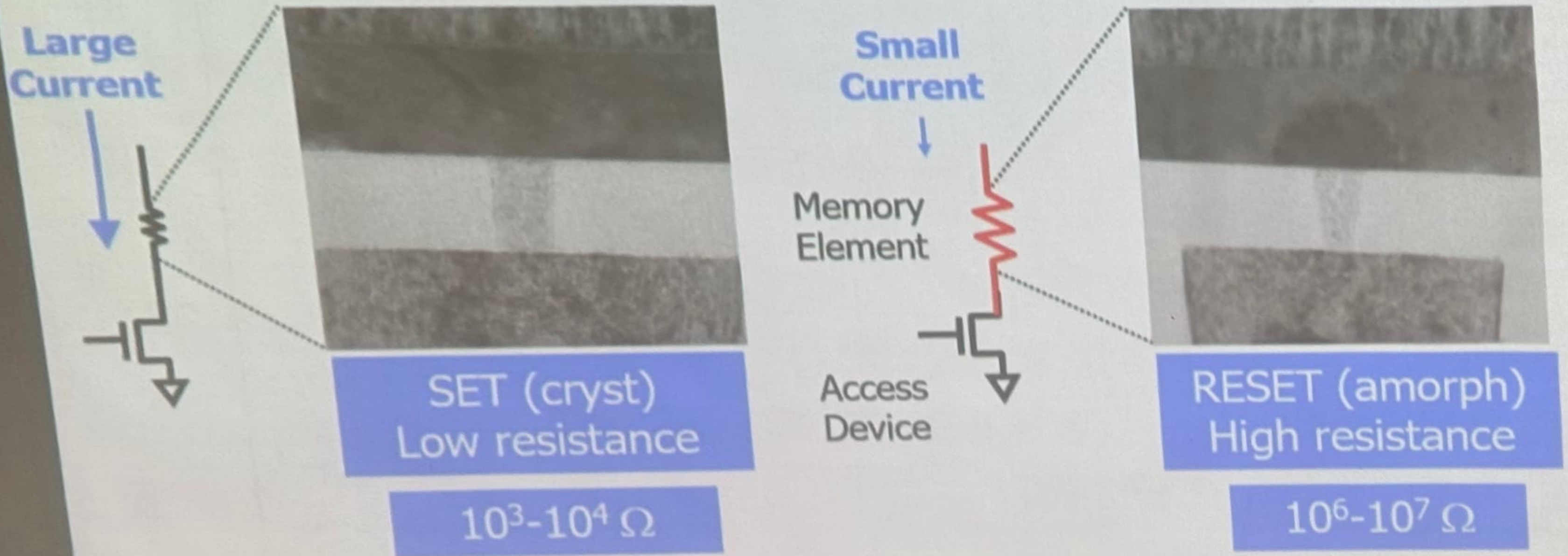


Photo Courtesy: Bipin Rajendran, IBM   Slide Courtesy: Moinuddin Qureshi, IBM

Opportunity: PCM Advantages

- Scales better than DRAM, Flash
  - Requires current pulses, which scale linearly with feature size
  - Expected to scale to 9nm (2022 [ITRS])
  - Prototyped at 20nm (Raoux+, IBM JRD 2008)

- Can be denser than DRAM
  - Can store multiple bits per cell due to large resistance range
  - Prototypes with 2 bits/cell in ISSCC' 08, 4 bits/cell by 2012

- Non-volatile
  - Retain data for >10 years at 85C

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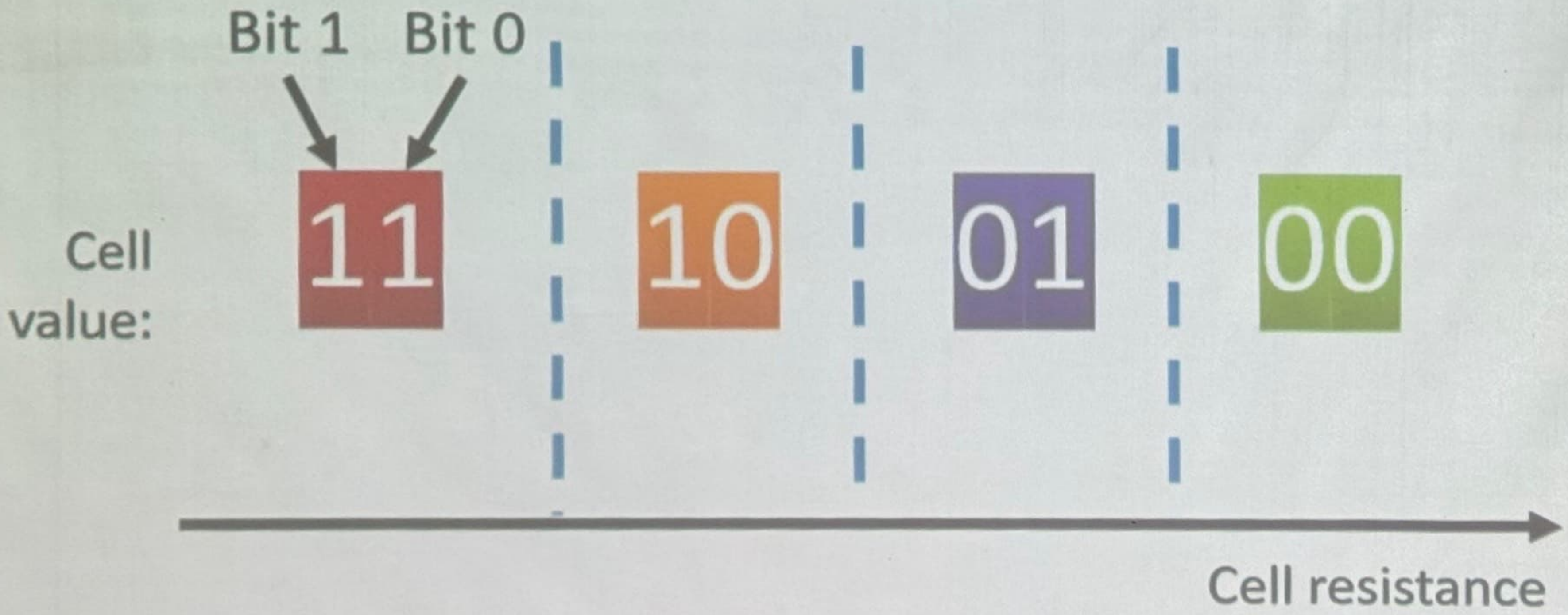
- Non-volatile
  - Retain data for >10 years at 85C

- No refresh needed, low idle power

MLC-PCM Resistance → Value

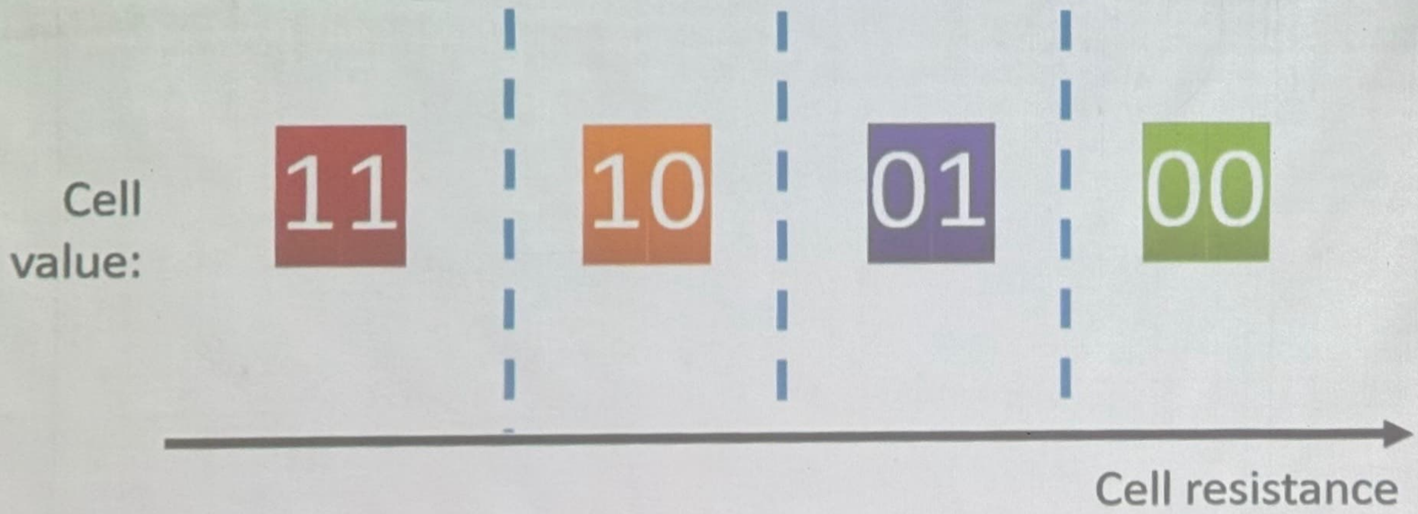


MLC-PCM Resistance → Value

Less margin between values

→ need more precise sensing/modification of cell contents

→ higher latency/energy (~2x for reads and 4x for writes)



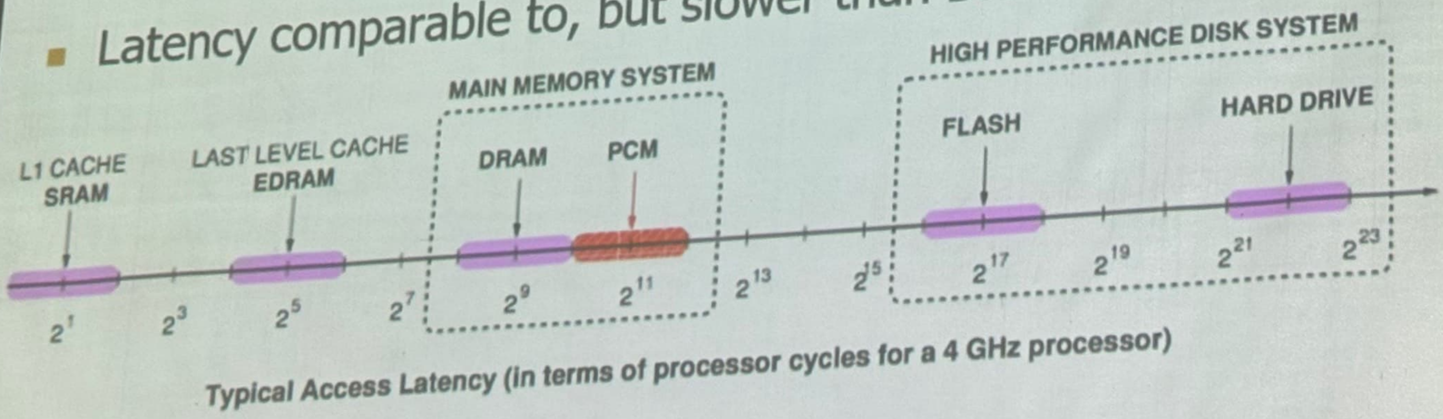
Phase Change Memory Properties

- Surveyed prototypes from 2003-2008 (ITRS, IEDM, VLSI, ISSCC)
- Derived PCM parameters for F=90nm

- Lee, Ipek, Mutlu, Burger, "Architecting Phase Change Memory as a Scalable DRAM Alternative," ISCA 2009.
- Lee et al., "Phase Change Technology and the Future of Main Memory," IEEE Micro Top Picks 2010.

Phase Change Memory Properties: Latency

- Latency comparable to, but slower than DRAM



- Read Latency
  - 50ns: 4x DRAM, 10^{-3}x NAND Flash
- Write Latency
  - 150ns: 12x DRAM
- Write Bandwidth
  - 5-10 MB/s: 0.1x DRAM, 1x NAND Flash

Qureshi+, "Scalable high performance main memory system using phase-change memory technology," ISCA 2009.

Phase Change Memory Properties

- Dynamic Energy
  - 40 uA Rd, 150 uA Wr
  - 2-43x DRAM, 1x NAND Flash
- Endurance
  - Writes induce phase change at 650C
  - Contacts degrade from thermal expansion/contraction
  - 10^8 writes per cell
  - 10^{-8}x DRAM, 10^3x NAND Flash
- Cell Size
  - 9-12F^2 using BJT, single-level cells
  - 1.5x DRAM, 2-3x NAND (will scale with feature size, MLC)

Phase Change Memory: Pros and Cons

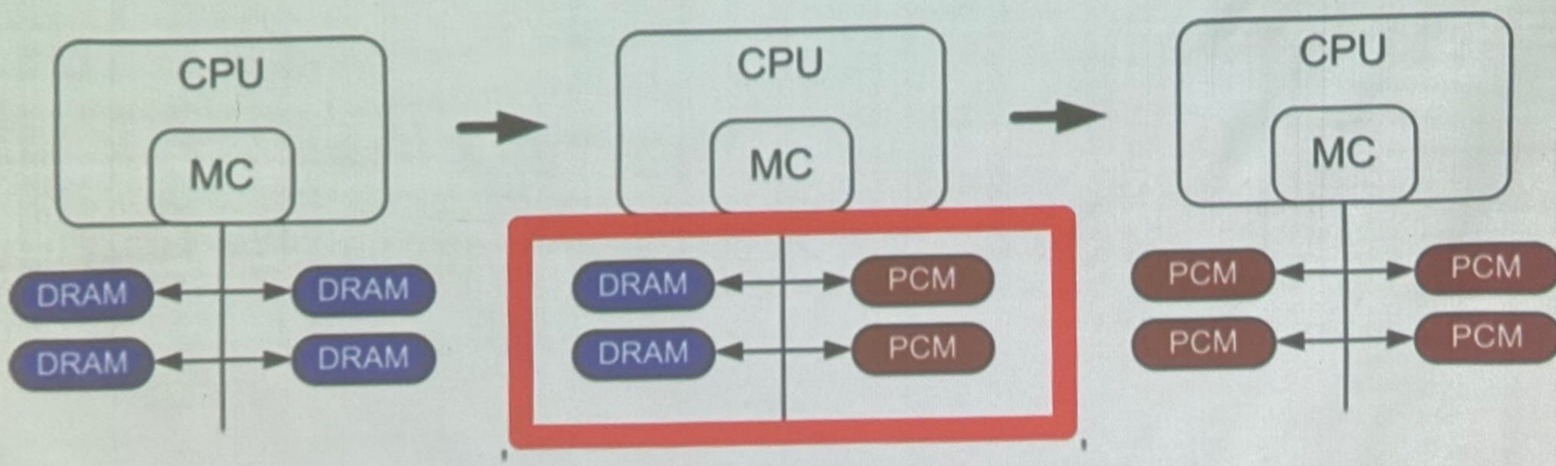
- Pros over DRAM
  - Better technology scaling (capacity and cost)
  - Non volatile → Persistent
  - Low idle power (no refresh)

- Cons
  - Higher latencies: ~4-15x DRAM (especially write)
  - Higher active energy: ~2-50x DRAM (especially write)
  - Lower endurance (a cell dies after ~10^8 writes)
  - Reliability issues (resistance drift)

- Challenges in enabling PCM as DRAM replacement/helper:
  - Mitigate PCM shortcomings
  - Find the right way to place PCM in the system

PCM-based Main Memory (I)

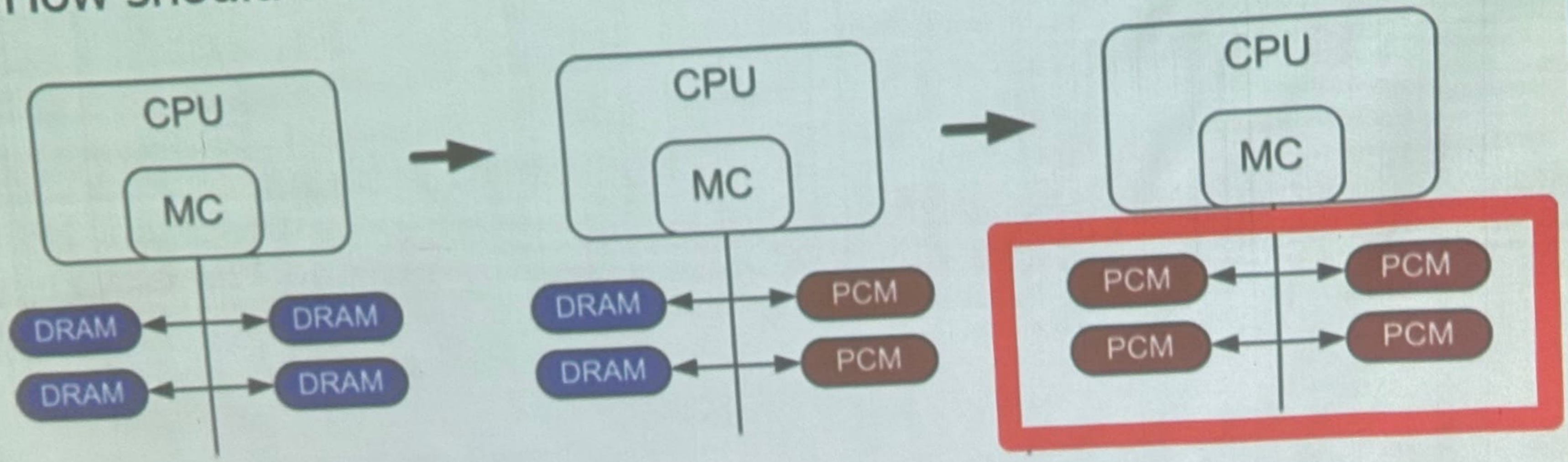
- How should PCM-based (main) memory be organized?



- Hybrid PCM+DRAM [Qureshi+ ISCA'09, Dhiman+ DAC'09]:
  - How to partition/migrate data between PCM and DRAM

PCM-based Main Memory (II)

- How should PCM-based (main) memory be organized?



- Pure PCM main memory [Lee et al., ISCA'09, Top Picks'10]:
  - How to redesign entire hierarchy (and cores) to overcome PCM shortcomings

An Initial Study: Replace DRAM with PCM

- Lee, Ipek, Mutlu, Burger, "Architecting Phase Change Memory as a Scalable DRAM Alternative," ISCA 2009.
  - Surveyed prototypes from 2003-2008 (e.g. IEDM, VLSI, ISSCC)
  - Derived "average" PCM parameters for F=90nm

Density
- ▷ 9 - 12F^2 using BJT
- ▷ 1.5× DRAM

Latency
- ▷ 50ns Rd, 150ns Wr
- ▷ 4×, 12× DRAM

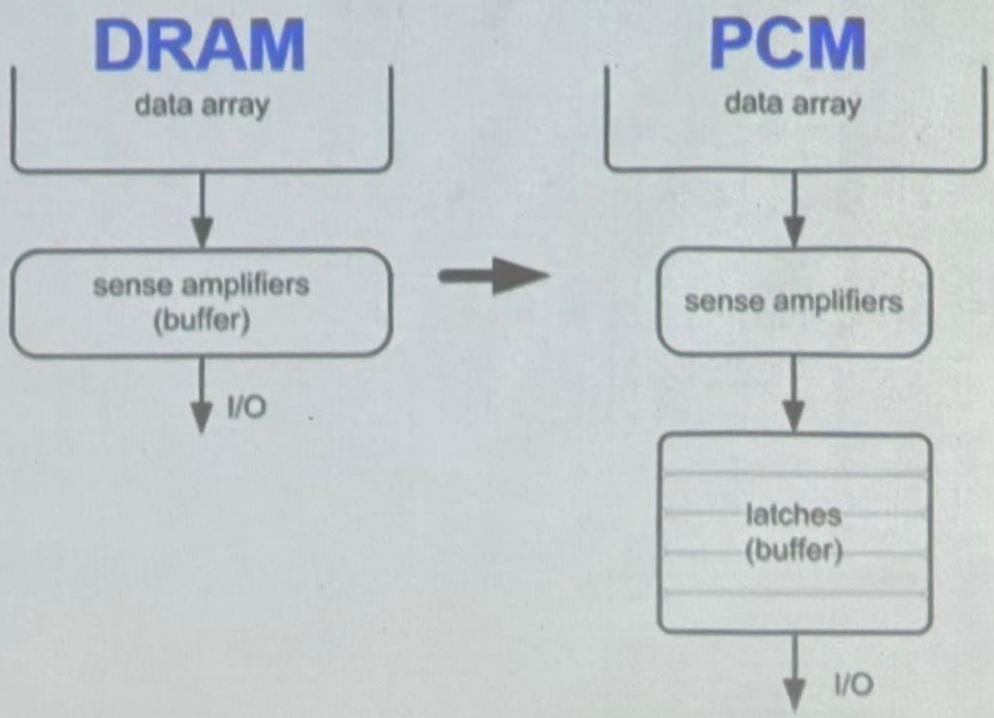
Endurance
- ▷ 1E+08 writes
- ▷ 1E-08× DRAM

Energy
- ▷ 40μA Rd, 150μA Wr
- ▷ 2×, 43× DRAM

Architecting PCM to Mitigate Shortcomings

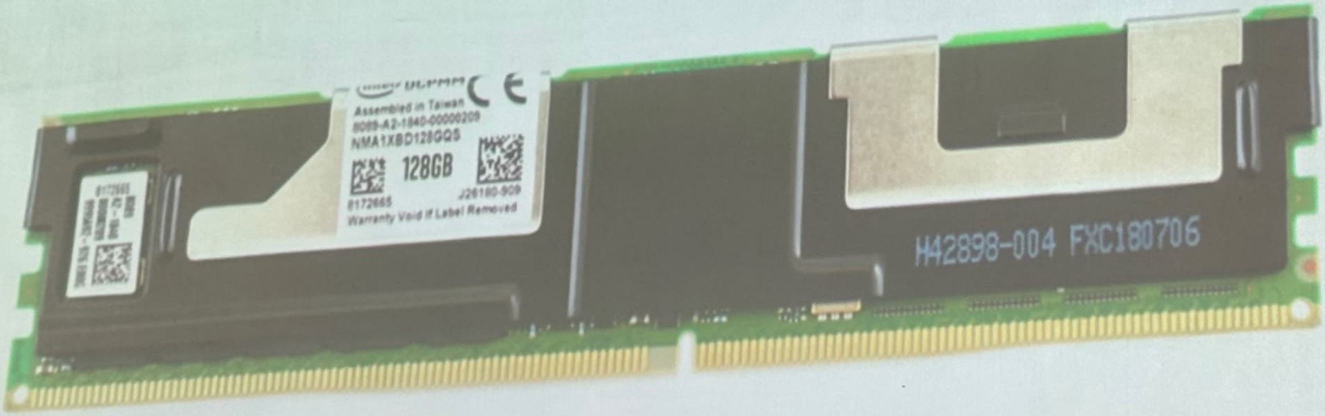
- Idea 1: Use multiple narrow row buffers in each PCM chip
  → Reduces array reads/writes → better endurance, latency, energy

- Idea 2: Write into array at cache block or word granularity
  → Reduces unnecessary wear



Intel Optane Memory (discontinued in 2022)

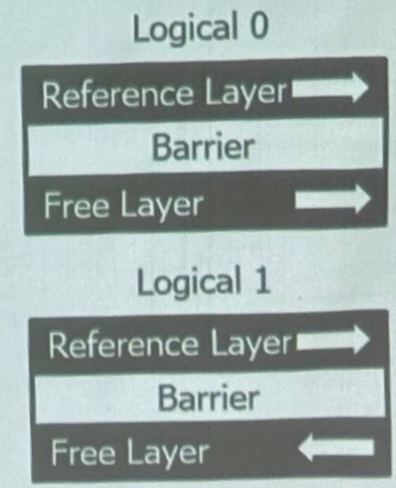
- Non-volatile main memory
- Based on 3D-XPoint Technology



https://www.storagereview.com/intel_optane_dc_persistent_memory_module_pmm

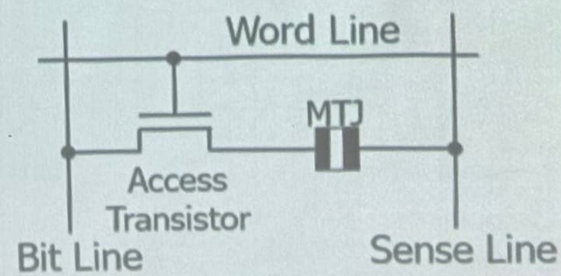
STT-MRAM as Main Memory

- Magnetic Tunnel Junction (MTJ) device
  - Reference layer: Fixed magnetic orientation
  - Free layer: Parallel or anti-parallel



- Magnetic orientation of the free layer determines logical state of device
  - High vs. low resistance

- Write: Push large current through MTJ to change orientation of free layer
- Read: Sense current flow



- Kultursay et al., "Evaluating STT-RAM as an Energy-Efficient Main Memory Alternative," ISPASS 2013.

STT-MRAM: Pros and Cons

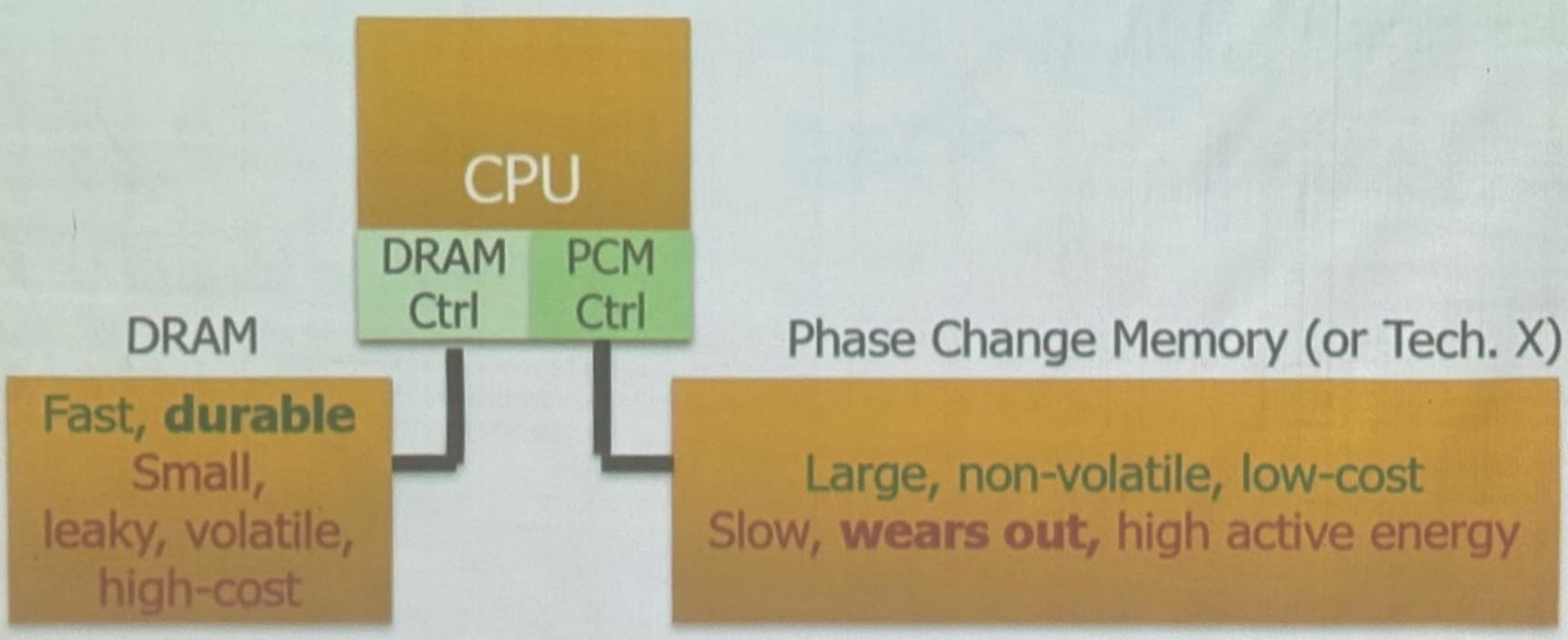
- Pros over DRAM
  - Better technology scaling (capacity and cost)
  - Non volatile → Persistent
  - Low idle power (no refresh)

- Cons
  - Higher write latency
  - Higher write energy
  - Poor density (currently)
  - Reliability?

- Another level of freedom
  - Can trade off non-volatility for lower write latency/energy (by reducing the size of the MTJ)

Hybrid Main Memory

A More Viable Approach: Hybrid Memory Systems



Hardware/software manage data allocation and movement
to achieve the best of multiple technologies

Meza+, "Enabling Efficient and Scalable Hybrid Memories," IEEE Comp. Arch. Letters, 2012.

Yoon+, "Row Buffer Locality Aware Caching Policies for Hybrid Memories," ICCD 2012 Best Paper Award.

Hybrid Memory Systems: Issues

- Cache vs. Main Memory
- Granularity of Data Move/Manage-ment: Fine or Coarse
- Hardware vs. Software vs. HW/SW Cooperative
- When to migrate data?
- How to design a scalable and efficient large cache?
- ...

One Option: DRAM as a Cache for PCM

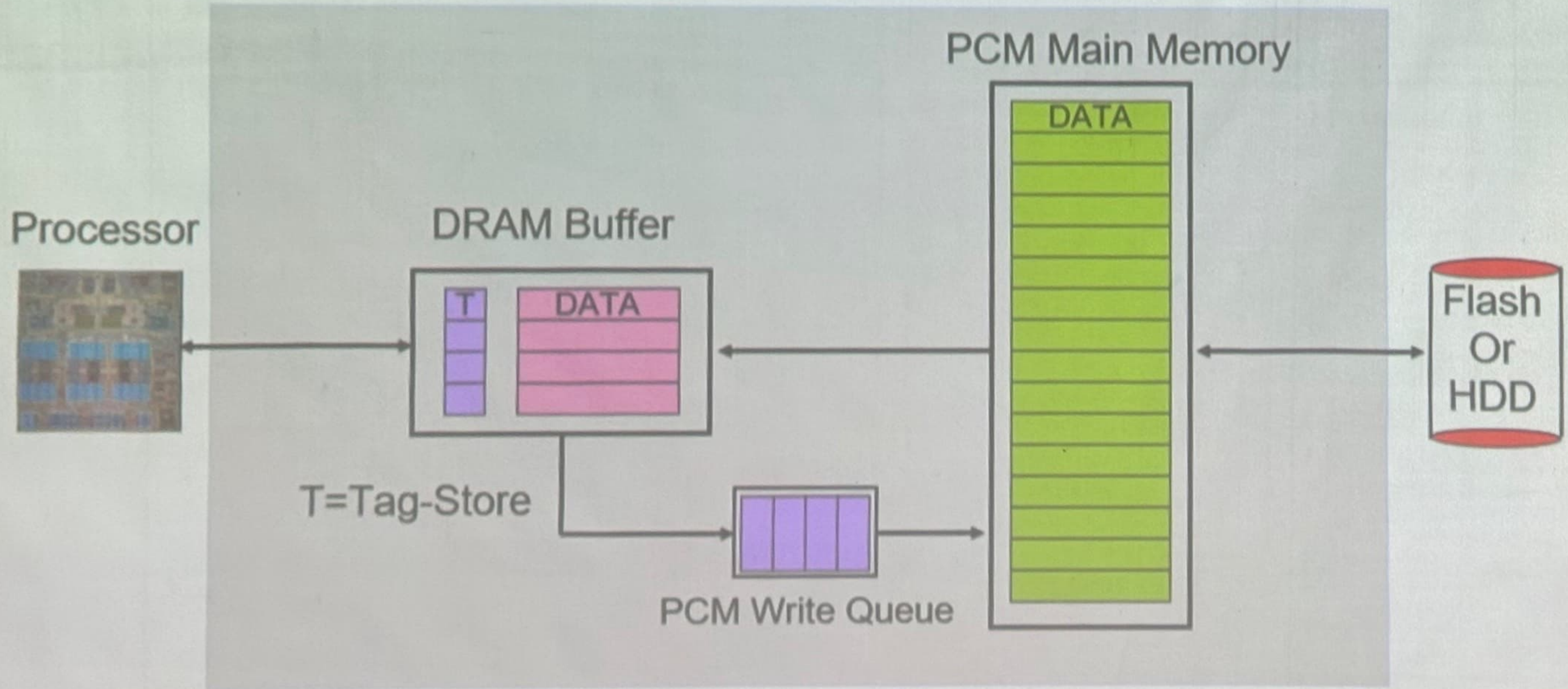
- PCM is main memory; DRAM caches memory rows/blocks
  - Benefits: Reduced latency on DRAM cache hit; write filtering
- Memory controller hardware manages the DRAM cache
  - Benefit: Eliminates system software overhead

- Three issues:
  - What data should be placed in DRAM versus kept in PCM?
  - What is the granularity of data movement?
  - How to design a low-cost hardware-managed DRAM cache?

- Two idea directions:
  - Locality-aware data placement [Yoon+ , ICCD 2012]
  - Cheap tag stores and dynamic granularity [Meza+, IEEE CAL 2012]

DRAM as a Cache for PCM

- Goal: Achieve the best of both DRAM and PCM/NVM
  - Minimize amount of DRAM w/o sacrificing performance, endurance
  - DRAM as cache to tolerate PCM latency and write bandwidth
  - PCM as main memory to provide large capacity at good cost and power



Qureshi+, "Scalable high performance main memory system using phase-change memory technology," ISCA 2009.

Other Opportunities with Emerging Technologies

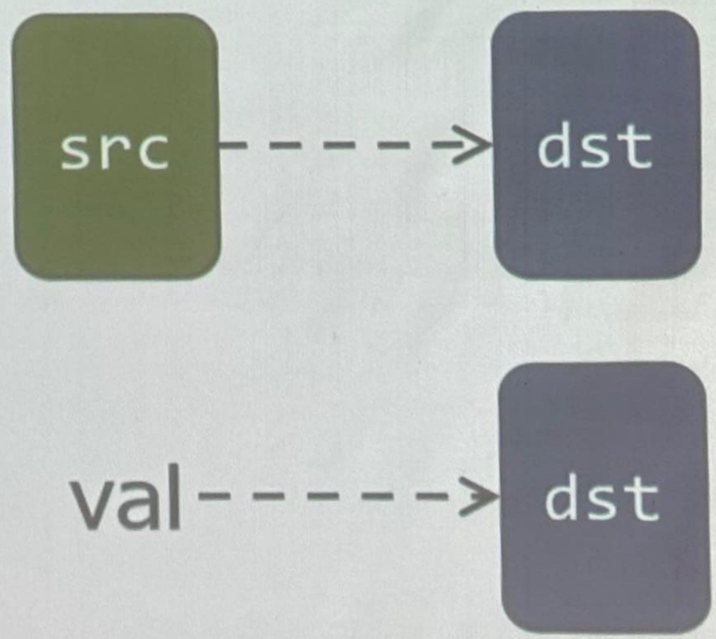
- Merging of memory and storage
  - e.g., a single interface to manage all data
- New applications
  - e.g., ultra-fast checkpoint and restore
- More robust system design
  - e.g., reducing data loss
- Processing tightly-coupled with memory
  - e.g., enabling efficient search and filtering

In-Memory Bulk Bitwise Operations

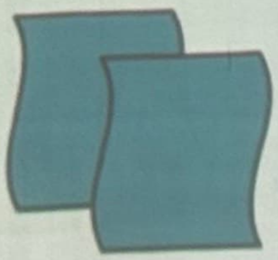
Starting Simple: Data Copy and Initialization

Bulk Data Copy

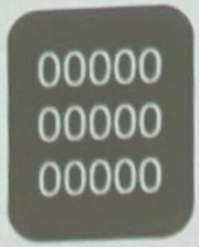
Bulk Data Initialization



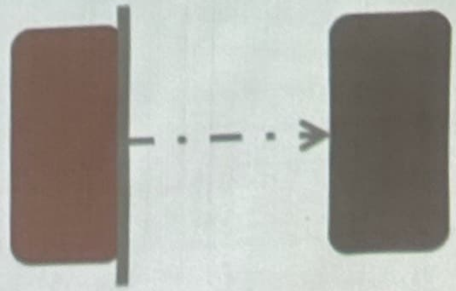
Starting Simple: Data Copy and Initialization



Forking

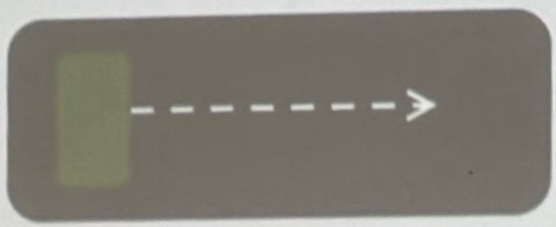


Zero initialization
(e.g., security)



Checkpointing

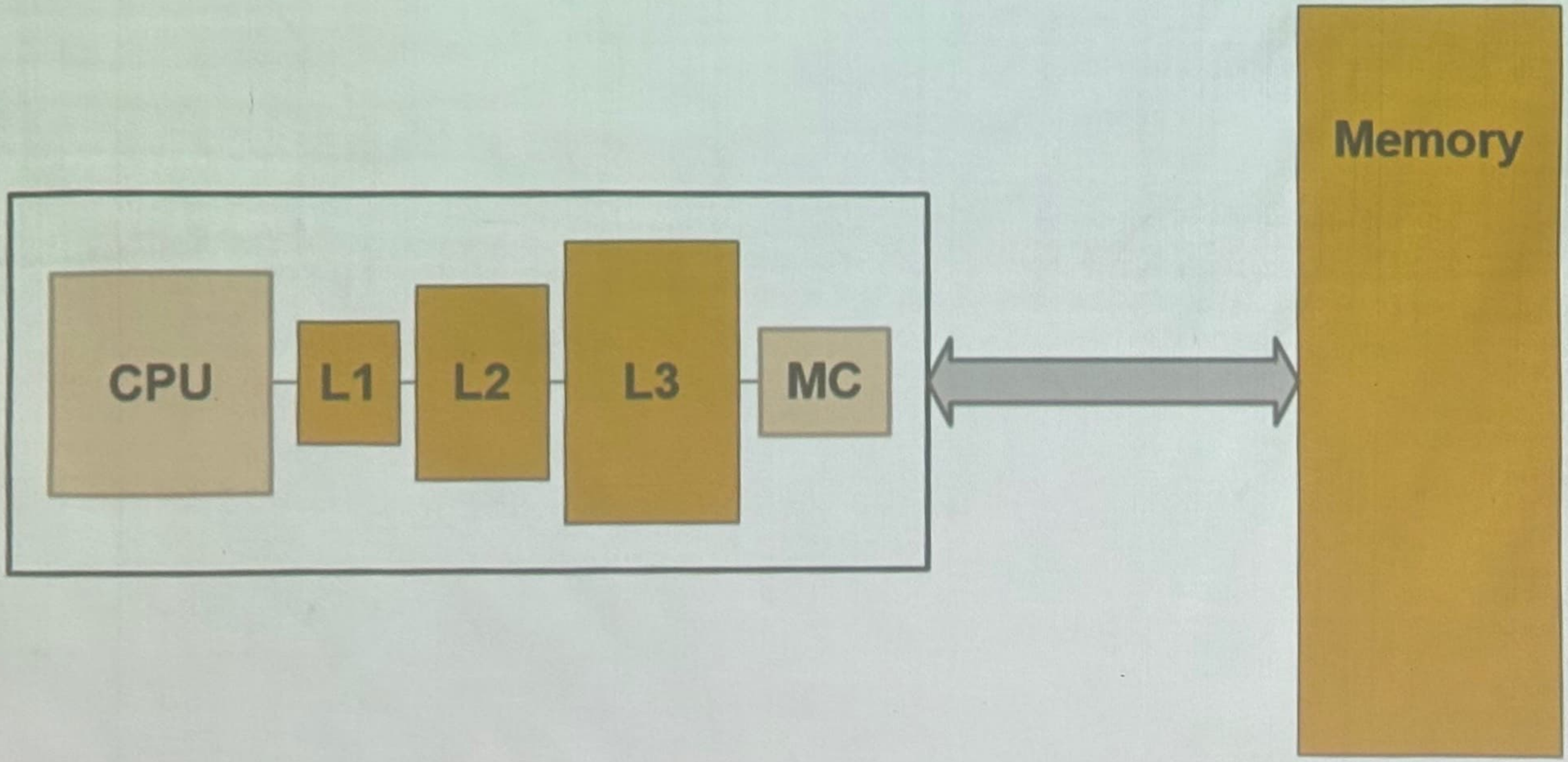
VM Cloning
Deduplication



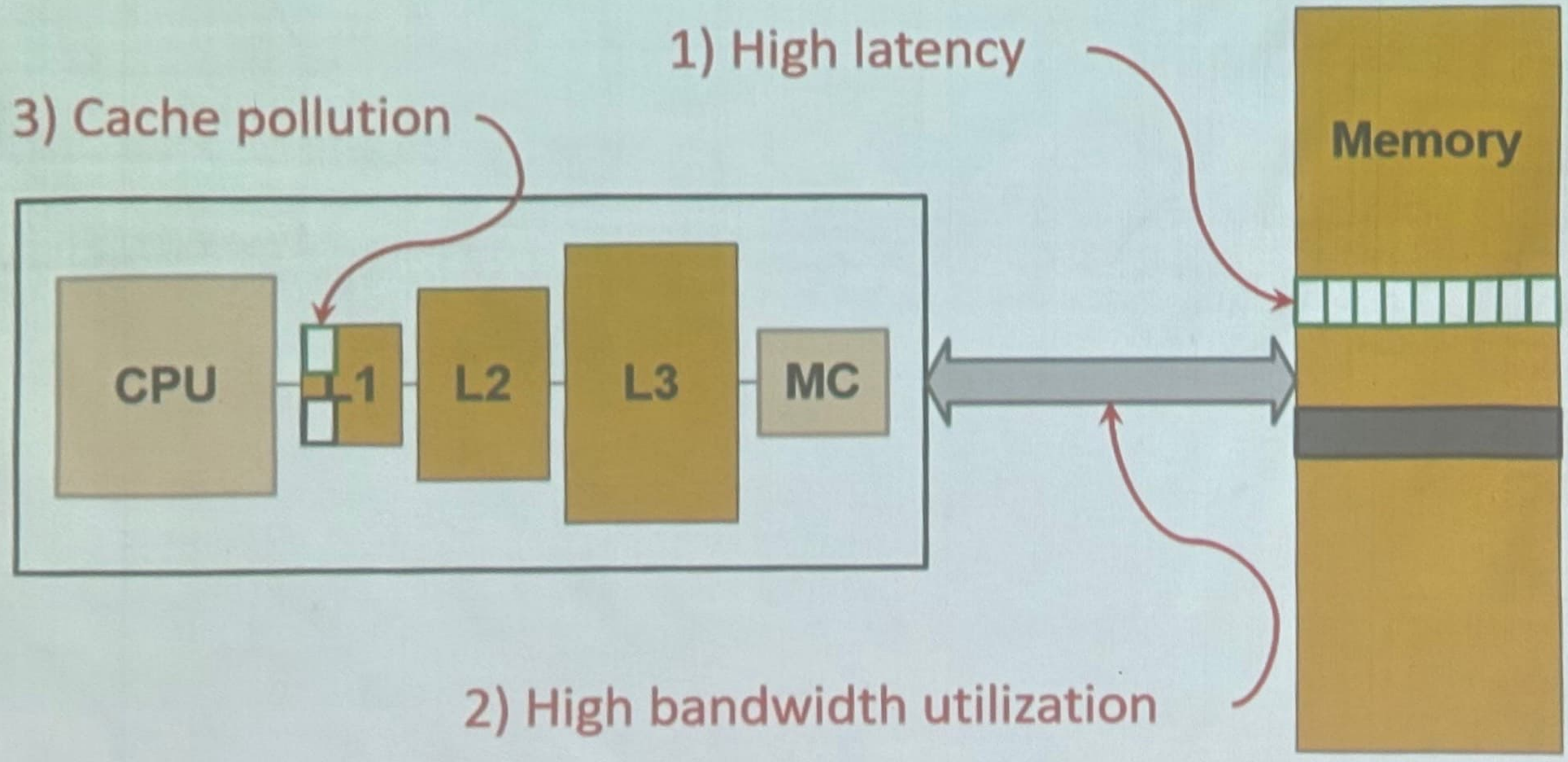
Page Migration

...
Many more

Today's Systems: Bulk Data Copy



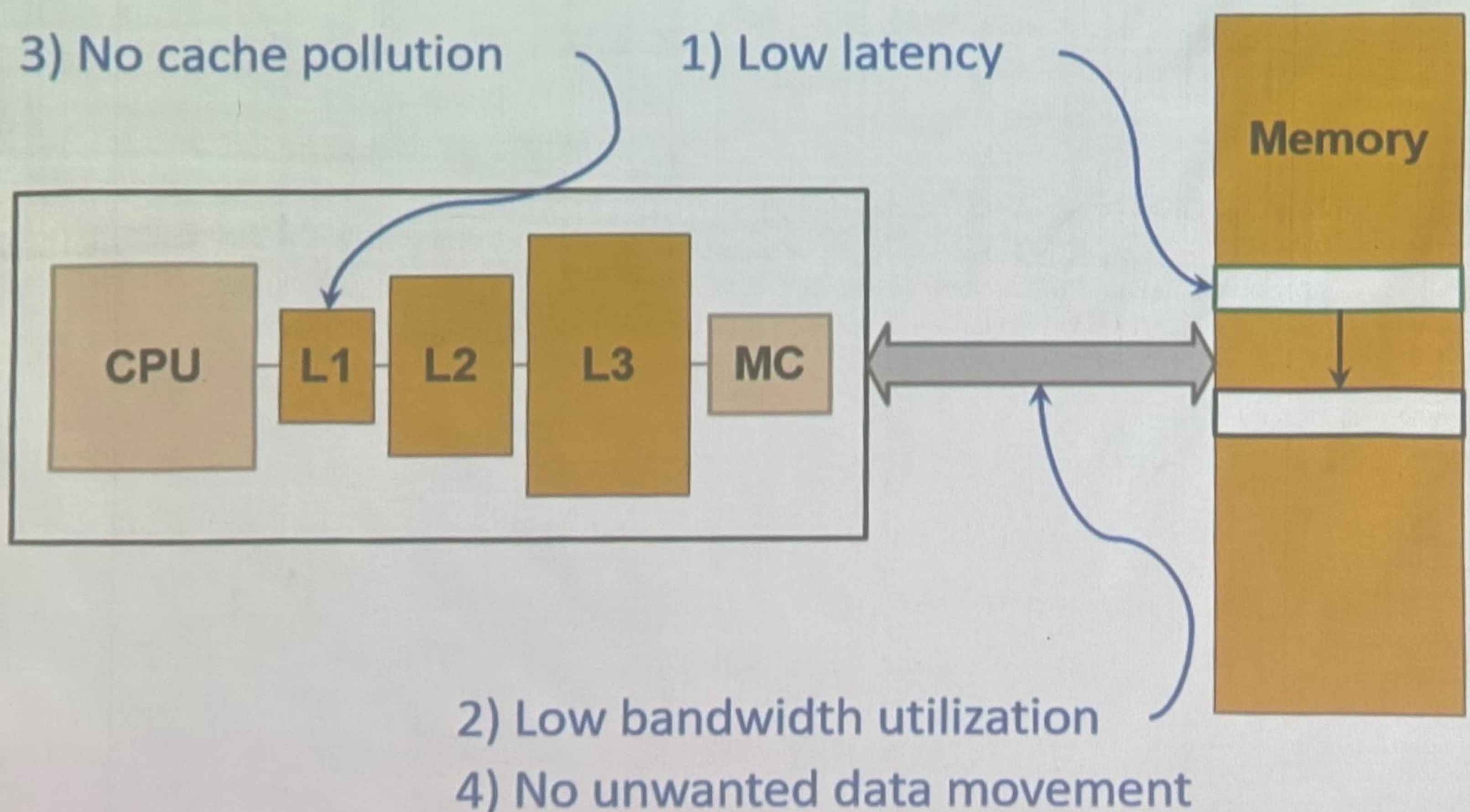
Today's Systems: Bulk Data Copy



4) Unwanted data movement

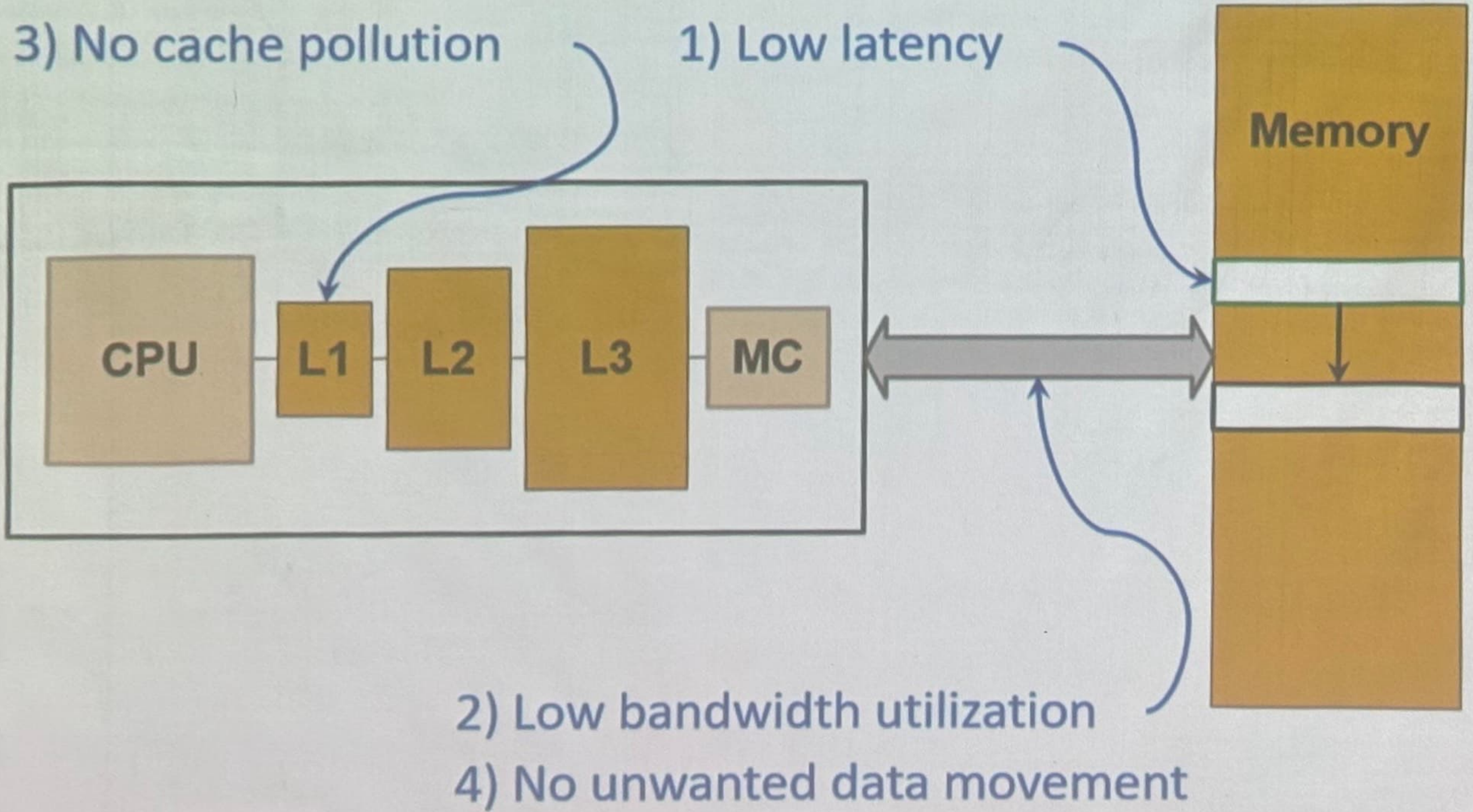
1046ns, 3.6uJ (for 4KB page copy via DMA)

Future Systems: In-Memory Copy



1046ns, 3.6uJ

Future Systems: In-Memory Copy



1046ns, 3.6uJ → 90ns, 0.04uJ

In-Memory Bulk Bitwise Operations

- We can support in-DRAM COPY, ZERO, AND, OR, NOT, MAJ
- At low cost
- Using analog computation capability of DRAM
  - Idea: activating multiple rows performs computation
- 30-60X performance and energy improvement
  - Seshadri+, "Ambit: In-Memory Accelerator for Bulk Bitwise Operations Using Commodity DRAM Technology," MICRO 2017.

- New memory technologies enable even more opportunities
  - Memristors, resistive RAM, phase change mem, STT-MRAM, ...

27-oct 25

- Most of architecture is dependent on memory

→ Memory intensive
— streaming workload (very little use of data)

then we have work loads which are

very large (with respect to data)
→ Comute [Also memory intensive]
cause of AI

way you look at it defines how comute change

what is Memory?
(in terms of physics)
→ must be something you are able to change / Retain

(Simplest Model)
Capacitance (C) → charge (q)

When we are Reading Memory → Sense charge (q)
when writing → change the capacitor

$$\underbrace{C|_{perfect} \rightarrow q}_{\text{No heat dissepation}}$$

there is no perfect capacitor, so there must be some Resistance
So voltage must be consumed after sometime

→ There are limit to building perfect capacitor, also limit to sensing we condo.

(very large) memory → less speed due to capacitance

As you increase size of Memory, sensing becomes difficult

Solution ①: New memory Architecture
⇝ Memory centric System design

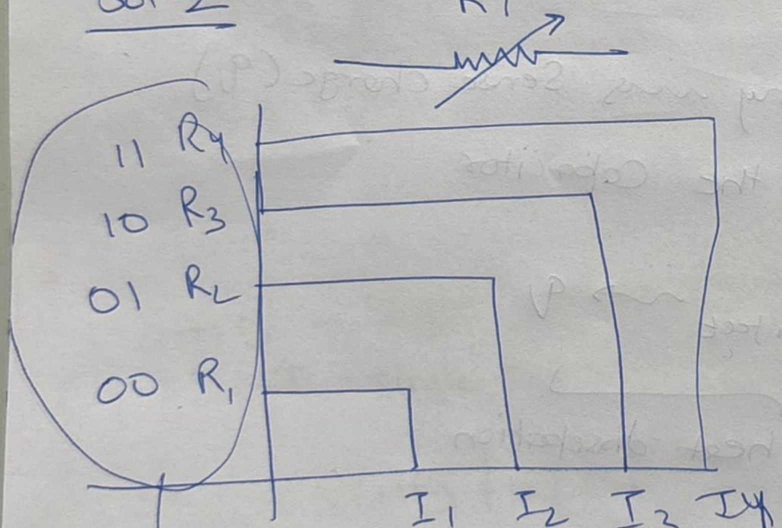
think of Memory behaviour Rather than Compute ~~befo~~ behaviour
Can my model ~~a~~ fit in memory,
its not about compute (more about fitting)

⇝ 1B parameter Model ⇝ 1B floating point NO
↓
Build implementation according to memory that you have ← fail with 1GPU

All this was in conventional memory (uses capacitor as basic element)

Sol-2

R → Programeble Res



Programming (write) currents

we will Read 1 of 4 Resistances

- Phase change memory
- 3D Memory
  - → How do you dissipate heat

(Phase change Memory)

→ How (PCM) works
  - → low Res - High current
  - → High Rest - low current

Can we Replace PCM with DRAM?

Heat sync Nowadays is part of ~~sp~~ Sync.

- Must ~~Reduce w~~ Reduce n/w

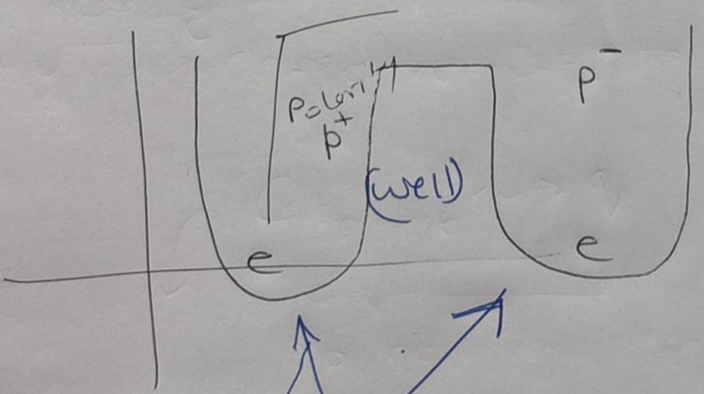
STT - MRAM

Magnetic Ram

~~Tunnelling?~~

Tunnelling

No energy give in Z dir
given in (x,y)



very close to eachother